Siddhesh Amrale

Data & AI Engineer | Forward-Deployed Builder

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SUMMARY

Engineer who builds data and AI systems alongside the people who use them, 6+ years. Started in full-stack product work in Pune, moved to mortgage data pipelines and APIs at Freddie Mac while earning an MS in Computer Science at NJIT, and now own Pittsburgh International Airport's data and AI platform. The loop: sit with the operations team, map the real workflow, ship a working version fast, measure it, then harden it into a platform. Shipped LLM enrichment, RAG agents with evaluation sets, streaming IoT pipelines and React apps staff use daily.

SKILLS

Build & ship: Python, TypeScript, React, Node.js, C#/.NET 8, Java, SQL/KQL, REST, GraphQL, Cypress, Rust

AI: LLM apps and pipelines, RAG, AI agents, MCP, LLM evaluation, Azure OpenAI, Semantic Kernel, Databricks ai_query, prompt versioning, MLflow

Data: Databricks (Lakeflow/DLT, Unity Catalog), Apache Spark, PySpark, Delta Lake, Airflow, Snowflake, Kafka, Event Hubs, Azure Data Explorer, ETL/ELT, CDC, data warehousing, data modeling, data quality, governance

Cloud & delivery: Azure, AWS (S3, Lambda, EKS), Terraform, Bicep, Docker, Kubernetes, GitHub Actions, Azure DevOps, ArgoCD, OIDC, Prometheus, Grafana, SLIs/SLOs

Working with customers: Workflow discovery with operations teams, rapid prototyping, production rollout, stakeholder management

EXPERIENCE

Analytics Full Stack Developer (Data & AI Platform Engineering) **Jan 2024 – Present**

Pittsburgh International Airport (Allegheny County Airport Authority), Pittsburgh, PA

Own the airport's data and AI platform end to end, from device and vendor feeds to the data products and apps operations staff use every day.

- Work directly with operations teams to map how they really work, then ship React + MSAL apps and Kusto-backed data APIs they use daily (parking kiosks, live gate status map, concessions, staff directory), covered by Cypress E2E tests. Incident response time down 40%.
- Built RAG agents on Azure OpenAI and Semantic Kernel over governed data, orchestrated with Airflow and checked against evaluation sets before changes ship. Five business teams now self-serve answers; ad-hoc report requests fell by about 60%.
- Put LLMs inside the pipeline: Databricks ai_query (Llama 3.3 70B) turns FAA NOTAMs and customer feedback into plain-English summaries and sentiment, with strict JSON output, versioned prompts, a kill switch and a content-hashed Delta cache that keeps reruns deterministic and cheap.
- Built a reusable ingestion platform on .NET 8 Azure Functions (16 function apps, ~50 connectors including Amadeus, SITA, Daifuku, Workday, TSA and FAA) with watermark/CDC loads, gap detection and time-range reprocessing. New sources now take 2 days instead of weeks; 1,300+ unit tests.
- Designed and built the One Data Platform on Databricks: 60 Lakeflow (DLT) pipelines and 58 jobs on Unity Catalog across flights, baggage, parking, ground transport, finance, workforce and customer experience. Auto Loader bronze, SCD2 silver and gold data products feed Power BI for 5 business units.
- Built the real-time path: ~70 Azure Functions and a Node.js decoder for LoRaWAN sensors, people counters and camera analytics, streaming through IoT Hub and Event Hubs into Azure Data Explorer (KQL) for occupancy, gate and parking data at about 5-minute freshness.
- Run the platform as code: Terraform, Databricks Asset Bundles, Bicep and Azure DevOps with OIDC and approval-gated production; Unity Catalog grants reconciled from code; alerts on pipeline failures, freshness and latency.
- Moved pipelines off Azure Data Factory onto serverless functions, identifying $400K+ a year in savings.

Software Engineer **Sep 2022 – Nov 2023**

Freddie Mac, New York, NY (remote)

Mortgage data pipelines and APIs for risk and compliance analytics, while completing an MS in Computer Science at NJIT.

- Built Python and SQL ingestion pipelines for loan-level mortgage data, with automated validation, reconciliation and quality checks that cut manual data review by 40%.
- Designed a GraphQL and REST data gateway (Node.js, Java) behind Azure API Management with Entra ID auth, caching and rate limiting, serving mortgage decisioning data at 1K+ requests/s with P95 latency under 10 ms.
- Defined SLIs, SLOs and error budgets, and added circuit breakers, idempotency and retries with backoff across data services, holding 99.95% uptime.
- Moved on-prem workflows and 12+ services onto Kubernetes (AKS, EKS) with GitHub Actions CI/CD and ArgoCD canary and blue-green releases: release cycles 50% shorter, deployment incidents down 30%.

Software Engineer, Full-Stack **Jan 2019 – Jul 2021**

Top Hat Software Technologies, Pune, India

Where I started: full-stack products and the data services behind them for enterprise clients.

- Built full-stack web applications (Angular, Java, MongoDB) for media content delivery over Azure Blob Storage and CDN, serving 10M+ requests a day at under 90 ms p99.
- Built batch ETL for content metadata, cutting refresh lag from 4 hours to under 20 minutes.
- Added Prometheus and Grafana dashboards and alerts on latency, throughput and error-rate SLIs, then fixed the worst bottlenecks: availability rose from 99.5% to 99.9%.

PROJECTS

Agent-Ready Lakehouse | Python, MCP, DuckDB, sqlglot, GitHub Actions | github.com/SiddheshAmrale/agent-ready-lakehouse
MCP server that gives AI agents governed, read-only access to lakehouse data: SQL allow-list, per-role sandboxes with masking and row filters, a locked engine and an audit log. A CI eval gate checks metric accuracy against an independent oracle, 23 attack and out-of-scope requests, and latency (p95 14 ms).

Lakehouse Forge | Python, DuckDB, Parquet, pytest | portfolio: /labs/forge
Runnable medallion lakehouse with CDC/SCD2, data-contract quarantine, late-arriving facts, freshness SLAs and experiment health checks.

VELA, native media player | Rust, libmpv, OpenGL | github.com/SiddheshAmrale/vela
Cross-platform (Windows, macOS, Linux) player that renders only when a frame or the UI changes: 5.6% CPU on HEVC playback vs 9.8% for stock mpv, 2.2 MB binary, 85 end-to-end checks.

Telemetry and incident diagnosis labs | Python, Linux, GitHub Actions | portfolio: /labs/telemetry, /labs/incident
Counter pipelines that hold up under duplicates, reordering, gaps and restarts; incident diagnosis scored on held-out cases, with live Linux network-namespace faults running in CI.

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (2025)
Microsoft Certified: Azure Developer Associate (AZ-204, 2024)
Microsoft Certified: Azure Administrator Associate (AZ-104, 2022)

EDUCATION

M.S. Computer Science, New Jersey Institute of Technology 2023
B.E. Computer Engineering, Savitribai Phule Pune University 2019